

AI Gmail Message Classifier

Google Apps Script | Gmail API | OpenAI Responses API | Strict Structured Outputs | Node.js verification

PLATFORM	AI CONTRACT	PRIVACY	QUALITY
Google Apps Script	Strict Structured Outputs	Minimized and redacted input	22 tests + multi-version CI

Architecture

INTAKE	MINIMIZATION	AI ADAPTER	EMAIL STATE
Individual unread inbox messages with bounded batch and run time.	Sender domain, redacted subject, generated snippet; no full body or attachment.	Pinned model, store: false, strict JSON schema, request IDs, bounded retries.	Namespaced category plus completion or review label; unrelated state remains intact.

Managed state

LABEL	PURPOSE
AI/Processed	Successful message-level classification
AI/Needs Review	Permanent provider or validation failure requiring review
AI/Invoice, AI/Order, ...	Allowlisted business category selected by validated output
AI/Other	Fallback for unknown or low-confidence classifications

Safety controls

- Untrusted email delimiters and prompt instructions	- Lock and runtime deadline
- Strict local schema and finite confidence	- Bounded retry and Retry-After support
- No secret-bearing status output	- Formula-safe spreadsheet values
- Explicit OAuth scope allowlist	- Content-free operational logging

Verification

22 tests plus repository and evaluation-fixture checks run on Node.js 20, 22, and 24. A dedicated Apps Script smoke-test function exercises one controlled provider request before trigger activation.

Run locally

npm install	npm run check	npm run clasp:status
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Production rollout includes scope approval, restricted credentials, provider budget controls, representative evaluation, and a dedicated Gmail smoke test.